

Arsalan Jawed, B.Sc.

Email: arsalan.jawed@outlook.com | +49 1709093258 | Hildesheim, Germany | [LinkedIn: Arsalan Jawed](#) | [GitHub](#) | Portfolio: arsalan-007.github.io

Personal Details

- Nationality: Pakistani
- Authorized to work in Germany (Student Visa)
- Language: English (fluent), German (A2) — several years of experience in UX/UI and web development

Relevant Skills

- Data Analysis (SQL, Python, Pandas, Matplotlib, Seaborn)
- Cloud & Warehousing (Databricks, Snowflake SQL, AWS S3, AWS EC2, AWS RedShift)
- Machine Learning and AI (Scikit-learn, PyTorch, MLLib)
- Web Stack (React.js/Redux, Javascript, Typescript, Java, MUI, SASS/SCSS, Tailwind)

Work Experience

Habib Bank AG Zurich

Oct 2024 - Mar 2025

Software Engineer

Karachi, Pakistan

- Led the development of an ETL pipeline for the wealth-management module, automating the ingestion, transformation, and validation of customer data and measurably improving data accuracy.
- Resolved production support tickets across multiple banking modules using SQL and Java, maintaining data integrity across core systems.
- Developed and integrated new features for Habib Bank AG Zurich's proprietary platform in Java, collaborating with cross-functional teams to deliver fixes on schedule.

Codex

Jun 2023 - Aug 2023

Front-End Developer

Karachi, Pakistan

- Developed a React Native Android messaging app with real-time chat, friend list management, and last-seen functionality.
- Built a web-based learning platform with an interactive code editor and integrated compiler for practicing programming languages.

Projects

ETL Pipeline Dashboard — Python, FastAPI, Pandas, scikit-learn, SQLAlchemy, React

- Built an end-to-end pipeline extracting live OHLCV data for 20 German equities, transforming it with Pandas (moving averages, RSI, volatility), and loading into SQLite via idempotent upserts to prevent duplication on overlapping runs.
- Trained a pooled gradient-boosted model to forecast next-day returns, validated walk-forward against a naive baseline to measure real predictive lift, surfaced live in a React dashboard driven by an APScheduler background job.

ML-TISS — Threat Intelligence & Scoring System — Python, XGBoost, scikit-learn,

FastAPI, React (B.Sc. Thesis)

- Classified network traffic into 10 attack categories on the UNSW-NB15 dataset (175K+ packets) using XGBoost, Random Forest, and Logistic Regression, with per-class metrics exposed through a FastAPI service.
- Mapped each detected threat to a CVSS v3.0 severity score with automated remediation guidance, rendered in an interactive React dashboard.

Real-Time Chat App — React, Node.js, Socket.io, MongoDB

- Built a full-stack messaging app with Socket.io WebSockets for instant delivery, typing indicators, read receipts, and live last-seen status persisted in MongoDB.
- Implemented bcrypt-hashed authentication with session persistence and a responsive React/Tailwind UI.

Education

Universität Hildesheim

April 2025 - Present

M.Sc. Data Analytics

Hildesheim, Germany

- Big Data Analytics: Multi-threading, Distributed Machine Learning, Scalable data storage and ETL
- Advanced Machine Learning: Decision Tree models, CNNs, RNNs, K-Means clustering
- Distributed Data Analytics Lab: Python, PySpark workloads, TensorFlow, PyTorch

Institute of Business Administration

Sep 2020 - Aug 2024

B.Sc. Computer Science

Karachi, Pakistan

- Courses: Calculus I, Calculus II, Linear Algebra, Artificial Intelligence
- Bachelor Thesis: "Enhancing Cyber Security through Machine Learning"